

Master's Comprehensive Examination

Part I

Theory (90 minutes)

March 10, 2001

Instructions: Answer three out of four questions. Open book and open notes.

1. Given the bivariate distribution in the accompanying table, find
 - (a) correlation (X, Y)
 - (b) $P(X = 1 | X + Y = 3)$
 - (c) $F_{X,Y}(2, 1)$, where $F_{X,Y}(x, y)$ is the joint cumulative distribution function of (X, Y) .
 - (d) $\text{Var}(Y | X = 1)$, i.e., the conditional variance of Y given $X = 1$.

		X	
		1	2
Y	0	.2	0
	1	.1	.2
	2	.3	.2

2. A bridge hand is a random selection of 13 cards from a standard deck of 52 playing cards.

- (a) Find the probability that a bridge hand contains at most one ace.
 - (b) If X is the number of aces in the hand, find EX and $\text{Var}(X - 9)$.
 - (c) Find the probability that a particular player receives no ace in each of the next four hands.

3. Let X_1, X_2 be a random sample of size 2 from a gamma distribution with parameters $\alpha = 3$ and θ . That is,

$$f_X(x; \theta) = \frac{e^{-x/\theta} x^2}{\Gamma(3)\theta^3}, \quad x > 0.$$

- (a) Find a sufficient statistic T .
 - (b) Is $\log_e(X_1 + X_2)$ a sufficient statistic?
 - (c) Find the maximum likelihood estimator of e^θ .

4. Let X_1, X_2, X_3 be independent $N(\mu, 1)$. Carry out the likelihood ratio test of size $\alpha = .05$ of $H_0 : \mu = 0$ vs $H_1 : \mu \neq 0$ given observations $X_1 = 1, X_2 = -1, X_3 = 3$.